



Design and Implementation of a Transduction System for the Measurement of a Weakly Nonlinear Mechanical Oscillator - Optical Proximity Sensor

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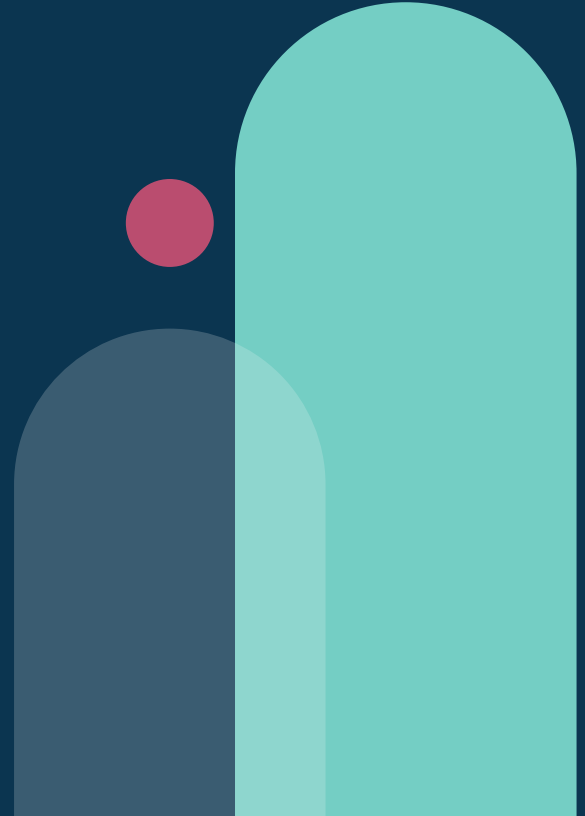
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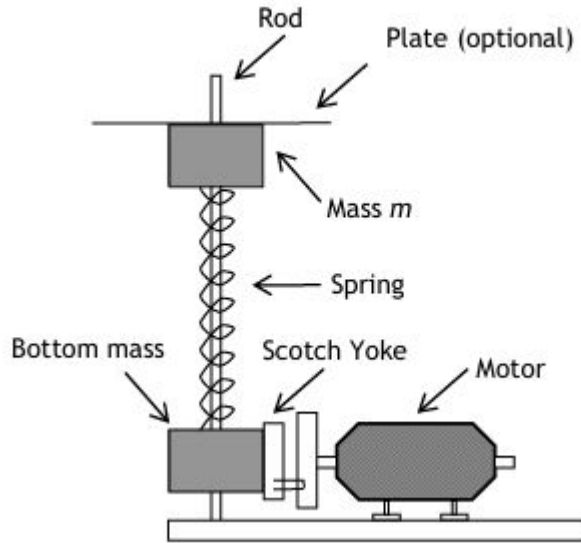


01

Theory



Oscillator Theory

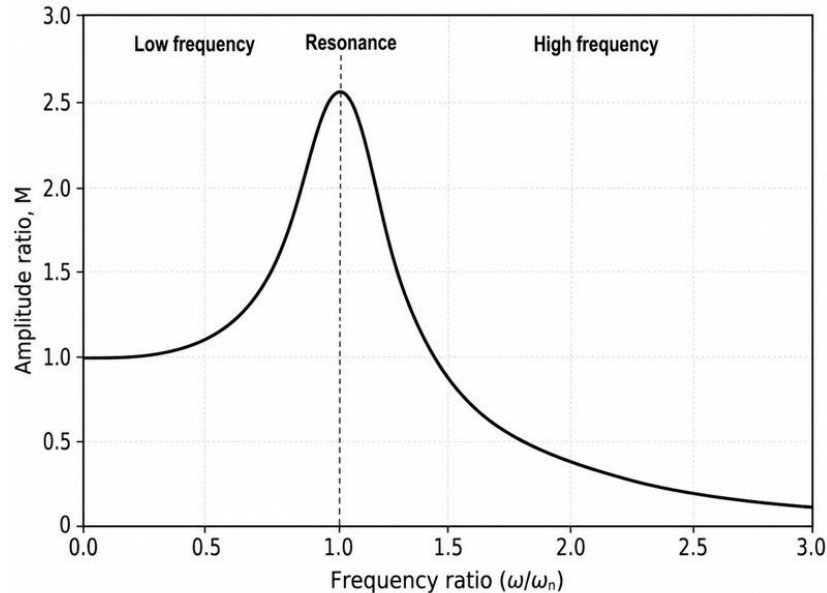


$$m\ddot{x} + c\dot{x} + kx = kX_b \sin(\omega t)$$

- The motor drives the bottom mass up and down. This motion is transferred through the spring to the top mass, but the top mass does not always move with the same amplitude as the bottom mass.
- The system behaves like a forced second-order oscillator. The response of the top mass depends on the driving frequency, spring stiffness, mass, and damping.

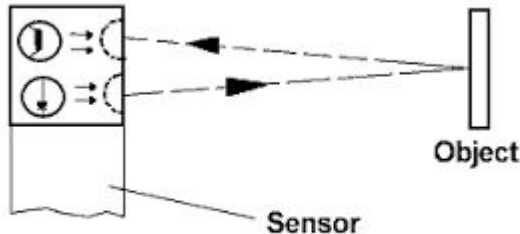
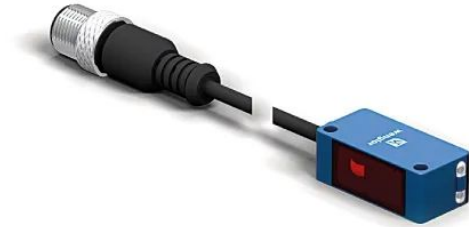
Oscillator Theory

$$M(\omega) = \frac{X}{X_b}$$



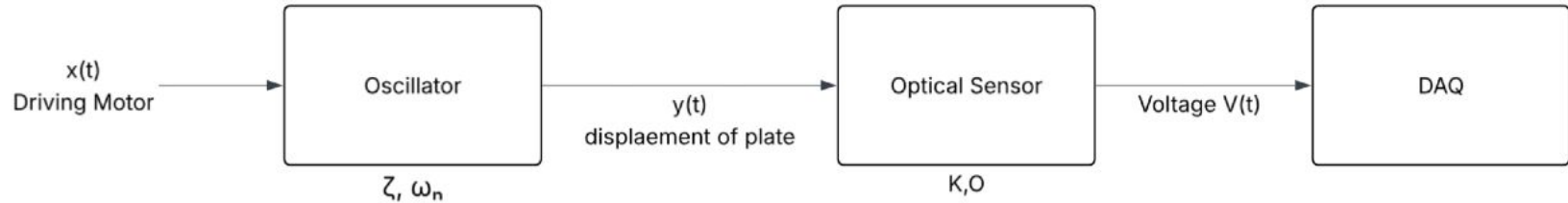
- At low driving frequency, the top mass follows the bottom mass closely.
- Near resonance, the top mass motion becomes amplified.
- At high driving frequency, inertia prevents the top mass from fully following the input motion.

Optical Proximity Sensor Description



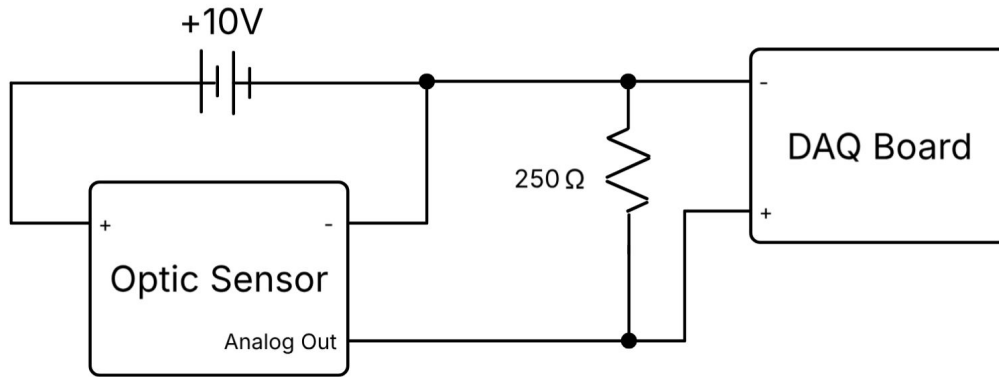
- Measures displacement without physical contact with the moving mass.
- The sensor emits continuous, modulated LED light toward the moving target surface.
- As the top mass oscillates, the target distance changes. The optical sensor detects this distance change through reflected light intensity and converts the motion into a time-varying electrical signal. After calibration, this signal represents the displacement of the top mass, $x(t)$.

Optical Proximity Sensor Block Diagram



- Motor drives displacement in the bottom mass which moves the top mass through a second order spring system
- Sensor measures displacement of top mass and outputs voltage which is sent to the DAQ

Optical Proximity Sensor Description



- Motor drive provides the input motion to the mechanical oscillator.
- Mechanical motion is transferred through the spring-mass-damper system.
- Optical proximity sensor measures the displacement of the moving mass without contact.
- Sensor output is sent to the voltage-divider signal conditioning circuit.
- Conditioned voltage signal V_{out} , is sent to the DAQ analog input.

02

Procedure





Procedure

Spring Constant Calibration

- Measure mass of bearing/plate, unloaded displacement
- Record displacements of plate under different loading values

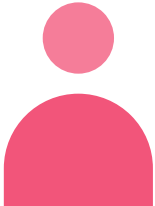
Sensor Configuration

- Turn on sensor, point at top plate
- Build circuit
- 10V input to sensor, 250 Ohm resistance, connect to MATLAB DAQ script

Sensor Calibration

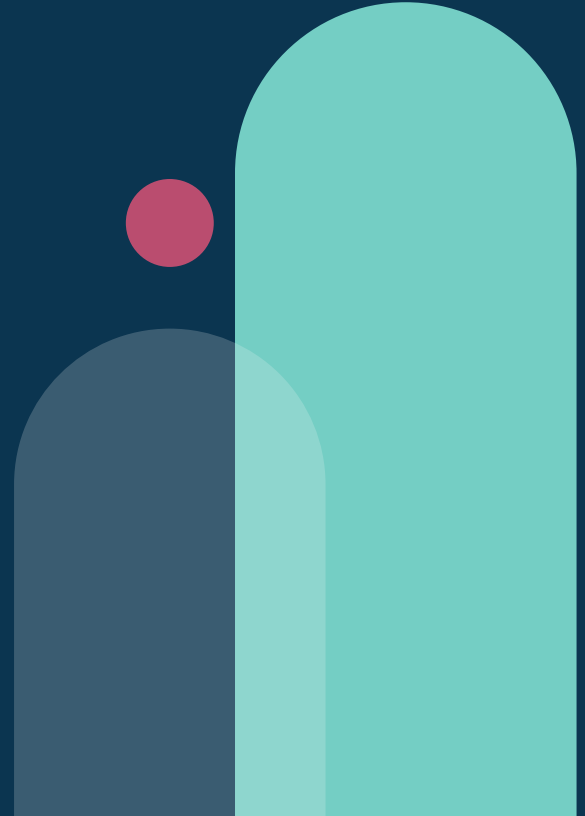
- Use script to observe and record voltage vs time dampening response curves for different **known** initial amplitude steps
- Repeat for many amplitudes

Sinusoidal Response

- Run script for different chosen driving frequencies
 - Change frequency after each trial
 - Account for hysteresis
- 

03

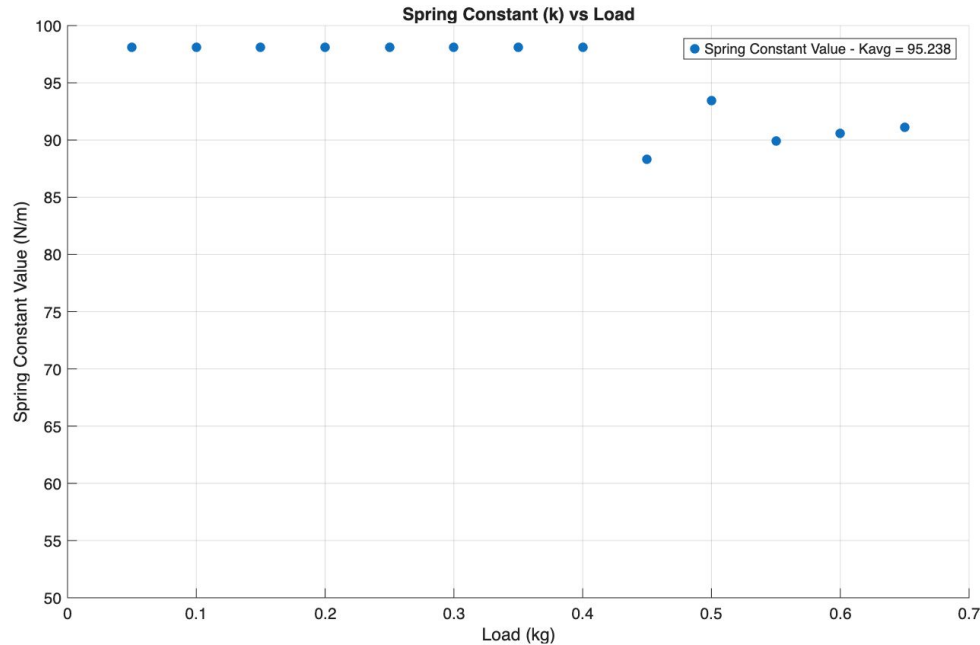
Analysis



Spring Constant

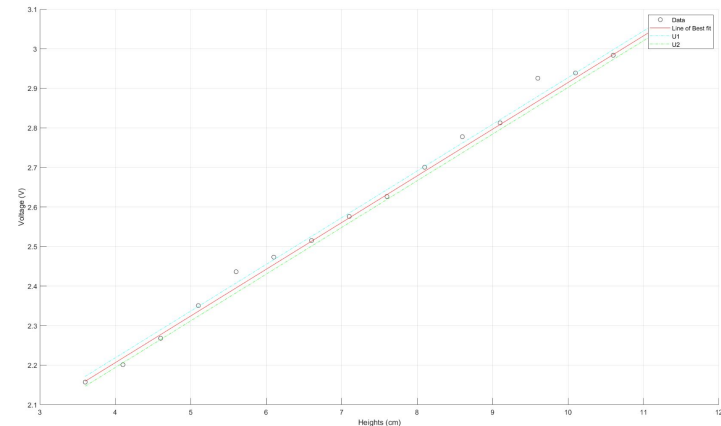
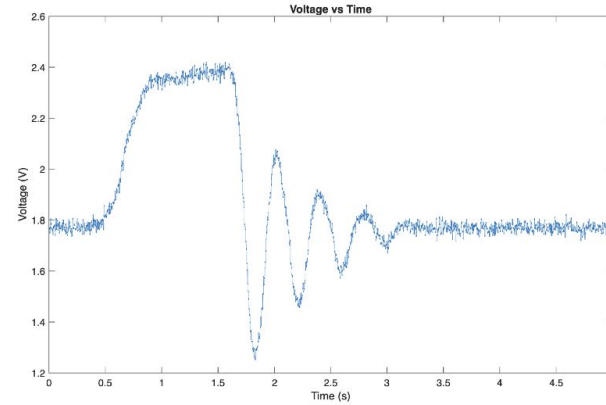
- Incremental masses were placed on the top plate, and their displacements were measured
- Hooke's Law states a direct proportionality between displacement and force needed for displacement

$$F = kx \rightarrow k = \frac{F}{x}$$



Sensor Calibration

- Zero-th order Sensor
 - No storage, no damping
 - Find K, O
- Executed using step response voltage output
 - Change DAQ script to 5 seconds
 - Run script, pull top plate down to a known mass, let go
 - Save file
- Obtain offset voltage and step voltage from the data
- Sensitivity: 0.1180 V/cm
- Offset: 1.735 V



Natural Frequency

Method 1: Second Order System Equation

- Known k from spring constant calcs
- m is the combined mass of the top plate and bearing
- **Natural Frequency = 19.3 rad/s**

$$\omega_n = \sqrt{\frac{k}{m}}$$

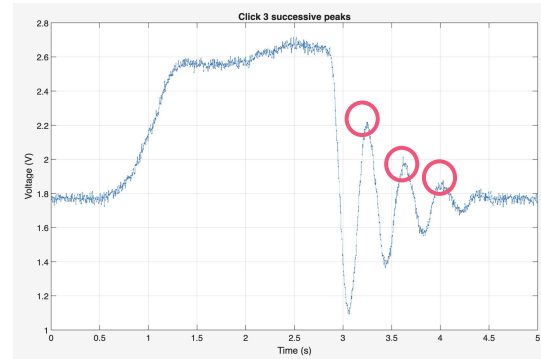
ω_n : natural frequency (rad/s)

k : spring constant (N/m)

m : mass (kg)

Method 2: Step Response Log Decrement

- Log decrement equation
- 'ginput' function in MATLAB
- Script calculates ζ , ω_d , ω_n
- **Natural Frequency = 16.67 rad/s**

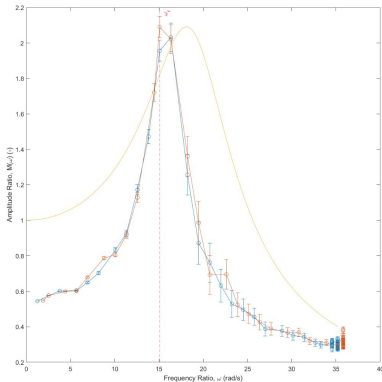


$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$
$$\omega_d = \frac{2\pi}{T_d}$$
$$\omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}}$$

Damping Ratio

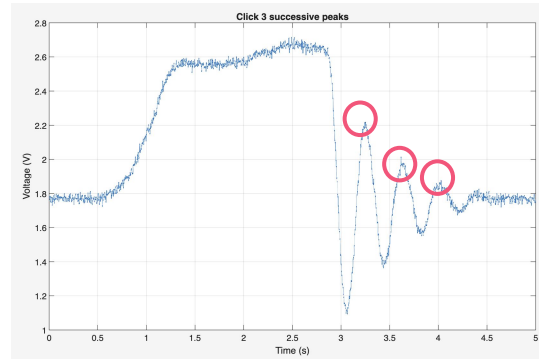
Method 1: Quality Factor Analysis

- Magnitude ratio at each driving frequency, identify maximum
- Set equal to quality factor and solve
- Script calculates M , ζ , ω , ω_r
- **Damping Ratio: 0.2468**



Method 2: Step Response Log Decrement

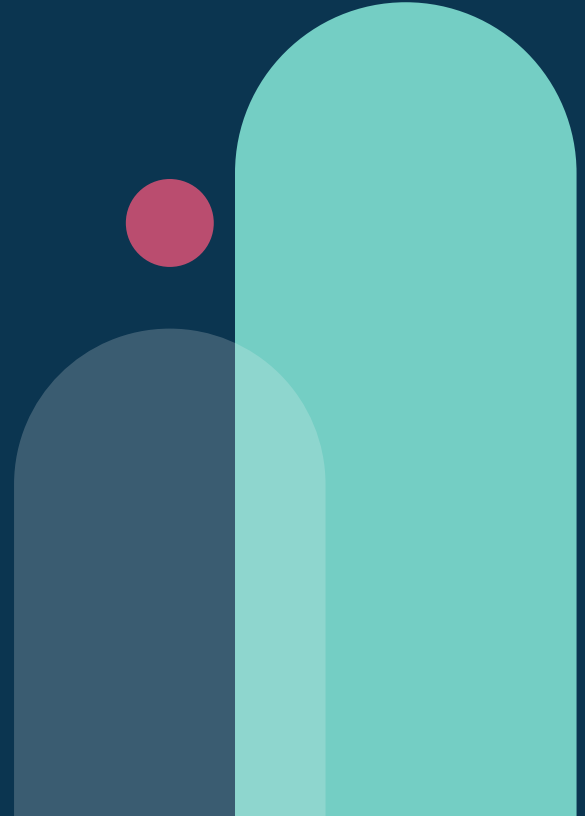
- Log decrement equation
- 'ginput' function in MATLAB
- Script calculates ζ , ω_d , ω_n
- **Damping Ratio = 0.0957**



$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$
$$\omega_d = \frac{2\pi}{T_d}$$
$$\omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}}$$

04

Uncertainty



Uncertainty Analysis

Spring Constant

- Random Uncertainty:
 $\pm 2.366 \text{ N/m}$
- Systematic Uncertainty:
 $\pm 1.27259 \text{ N/m}$
- Total Spring Constant
Uncertainty:
 $\pm 2.6904 \text{ N/m}$

Natural Frequency

- Uncertainty of the
Natural Frequency:
 $\pm 0.2731 \text{ rad/s}$

$$U_{\omega_n, \text{total}} = \frac{1}{2\sqrt{km}} U_{k, \text{total}}$$

Critical Damping Coefficient

- Uncertainty of the
Critical Damping
Ratio:
 $\pm 0.1634 \text{ N*s/m}$

$$U_{c_c, \text{total}} = \sqrt{\left(\sqrt{\frac{m}{k}} U_{k, \text{total}}\right)^2 + \left(\sqrt{\frac{k}{m}} U_m\right)^2}$$

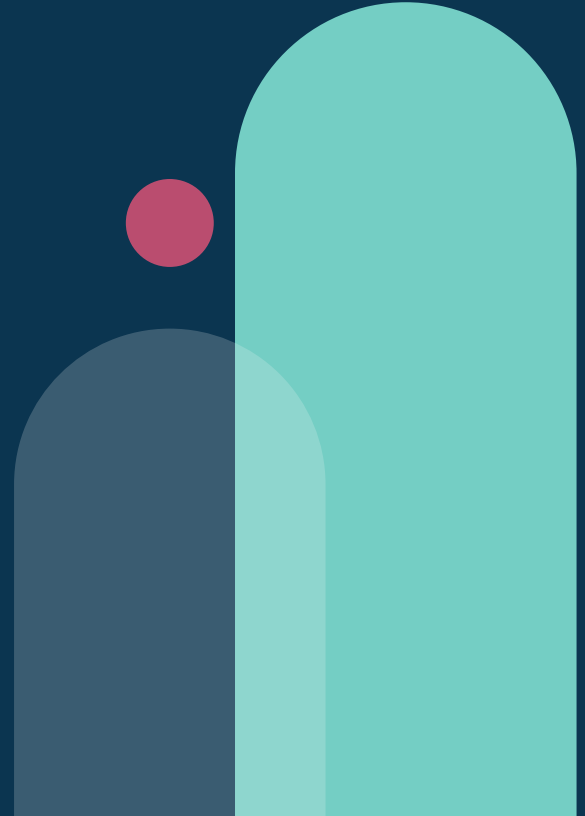
Damping Ratio

- Random Uncertainty:
 ± 0.0059
- Systematic
Uncertainty: ± 0.0141
Total Damping Ratio
Uncertainty: ± 0.0153

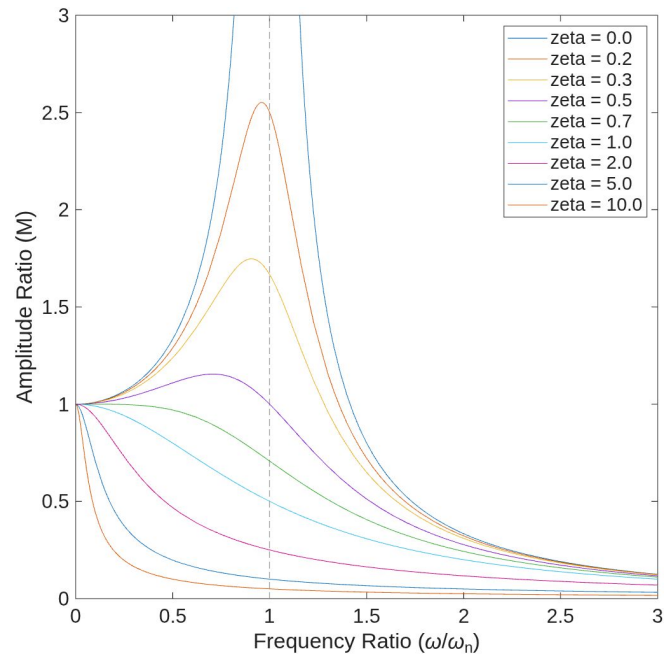
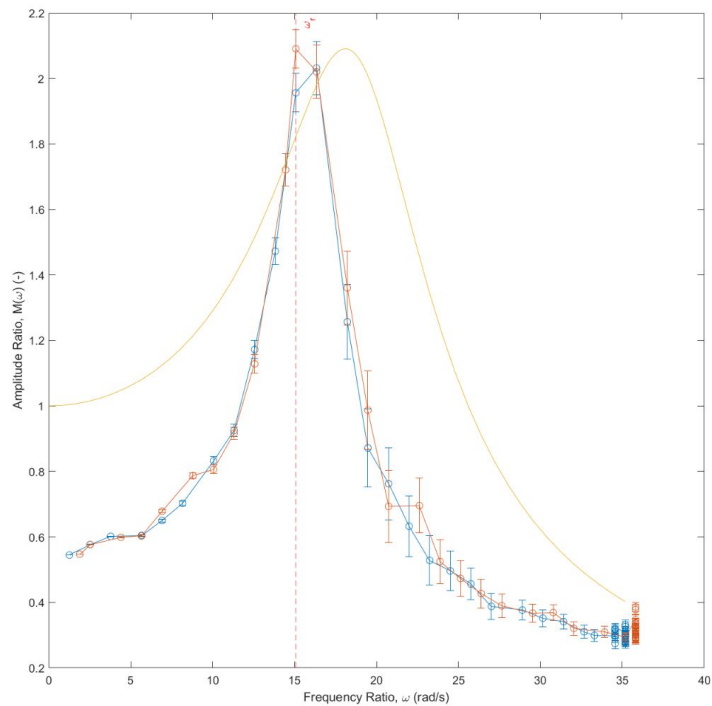
$$U_{\zeta} = \sqrt{\left(\frac{1}{\omega_n} * U_{w_n, \text{total}}\right)^2}$$

05

Results

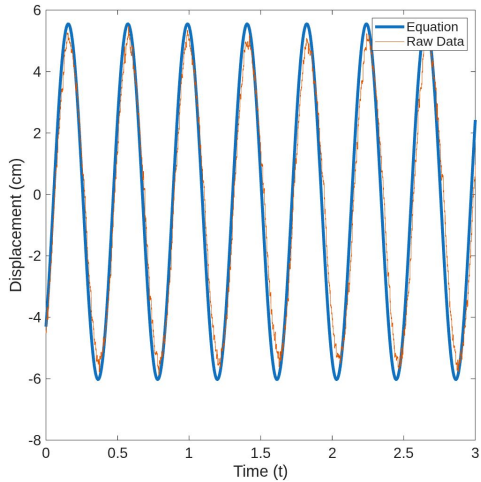


Magnitude Plot

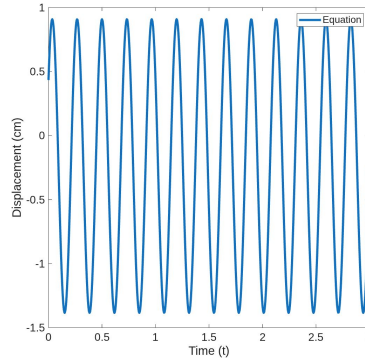


Representative Displacement Plots

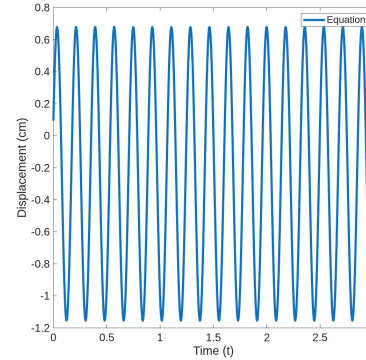
Trial 11:
Resonance Frequency Plot



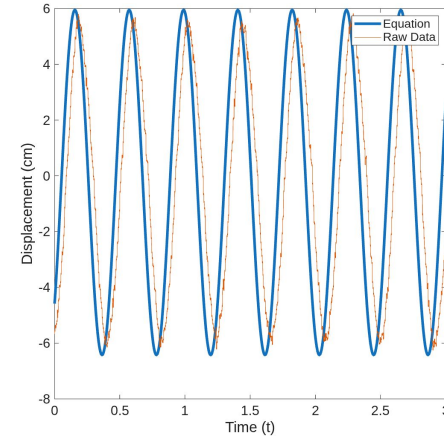
Trial 20



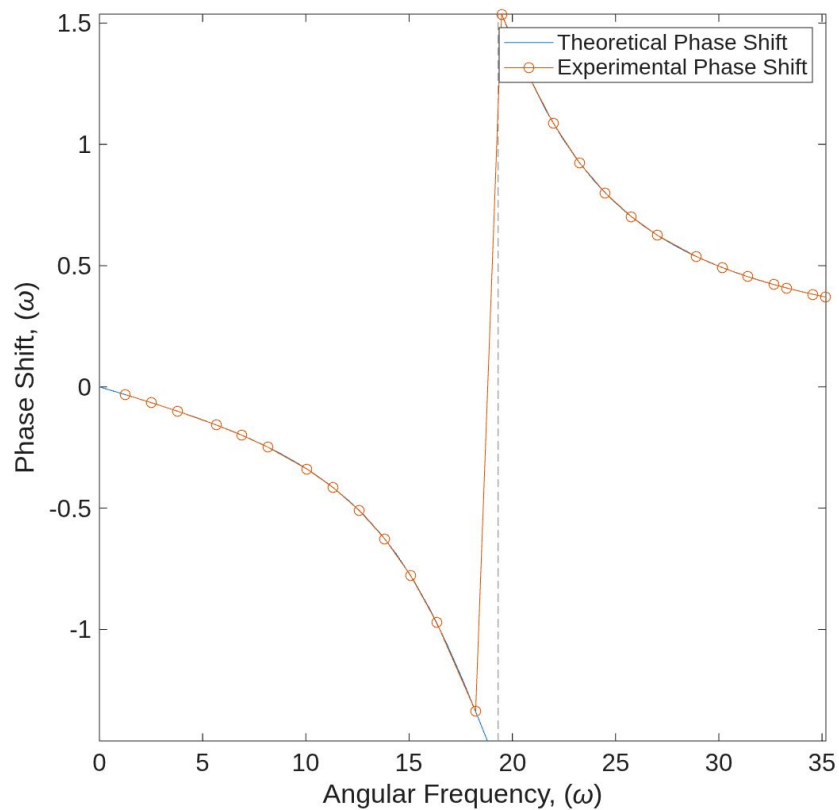
Trial 40



Trial 82



Phase Shift

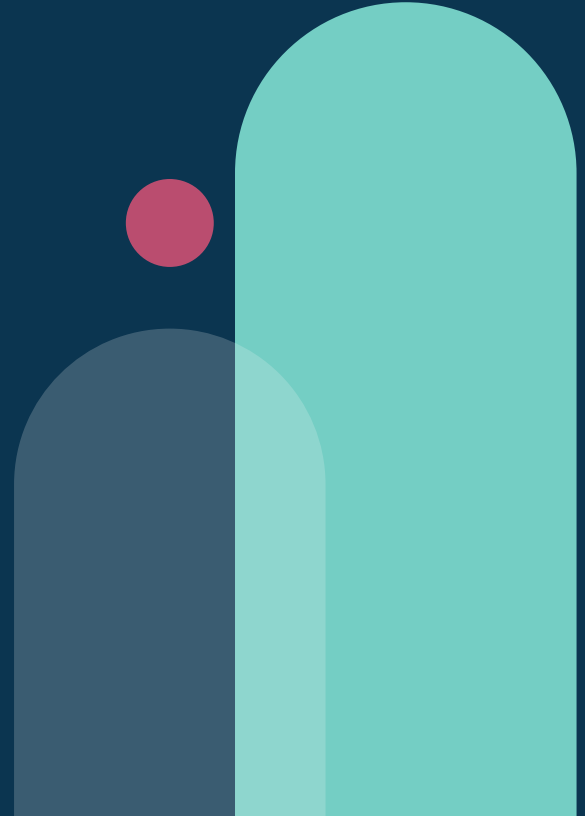


Final Results

Parameter	Variable Name	Value	Uncertainty	Unit
Spring Constant	k	95.238	± 2.6904	N/m
Natural Frequency	ω_n	16.67	± 0.2731	rad/s
Damping Ratio	ζ	0.2468	± 0.0153	-
Damping Coefficient	c	0.9432	± 0.014	$\frac{Ns}{m}$
Mass	m	0.255	± 0.0005	kg

06

Discussion/ Conclusion



Discussion and Conclusion

- Potential Sources of Error:
 - No spring mass consideration
 - No mass location consideration
 - Potential Spring Buckling
 - Calibration mistakes
- System still behaved predictably
- Shows sensor is consistent with distance measurements in the tested range
- Successfully modeled response of a second order system

Thank you!

